

KIBA-D

AI in scientific work and monitoring: LLMs for continuous biodiversity assessments

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Institute for Applied Ecology

UfU Unabhängiges Institut für Umweltfragen

AI in scientific work: mostly analysis of images and sound

Creation and analysis of big data



Camera traps

Species identification in millions of images



Bioacoustics

Birds, bats, insects in sound recordings



Remote sensing

Habitat change from satellite and drone data

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But:
much of what we know lies in text

- For most of history, findings were published as text only – original data often lost
- Monitoring data cover mostly recent decades
- They rarely explain drivers of change or the effect of measures

Large language models (LLM) can read digitized literature on a large scale

Germany‘s biodiversity assessment: a huge collective effort



145

authors

3.5

years of work

>1,200

pages of report

4,757

literature sources evaluated

More and more publications

+3,769

new publications in 3.5 years – on biodiversity & ecosystem services in Germany alone

Repeating this manually on a regular basis is very challenging

Sources: Faktencheck Artenvielfalt (Wirth et al. 2025); Web of Science search July 2022 vs. January 2026

How is an assessment made – and where can AI help?

Find literature

Search queries, screening of titles and abstracts, check for relevance

Read & annotate

Habitat, species, methods, drivers, measures …

Analyse trends

16,208 trend statements → vote counts per habitat and species group

Report & review

Synthesis, consensus, peer review



KIBA-D goals: Evaluate the AI potential in the assessment process

1. Literature search

- Crawler to retrieve literature
- Trained semantic filter

2. Text annotation

- LLM-pipeline for named entity recognition (NER)
- Text annotation to Faktencheck literature database

3. Trend identification

- LLM-pipeline for relation extraction (RE)
- Find biodiversity trends

Expand the corpus, towards repeated assessments

Evaluate AI output against Faktencheck data as a reference

Interdisciplinary team

DFKI: AI & language technology
iDiv: biodiversity, Faktencheck authors
UfU, Öko-Institut: law & data governance

Research within the FEdA initiative, 07/2025–03/2027

Faktencheck serves as reference

Literature lists

Results of systematic keyword search

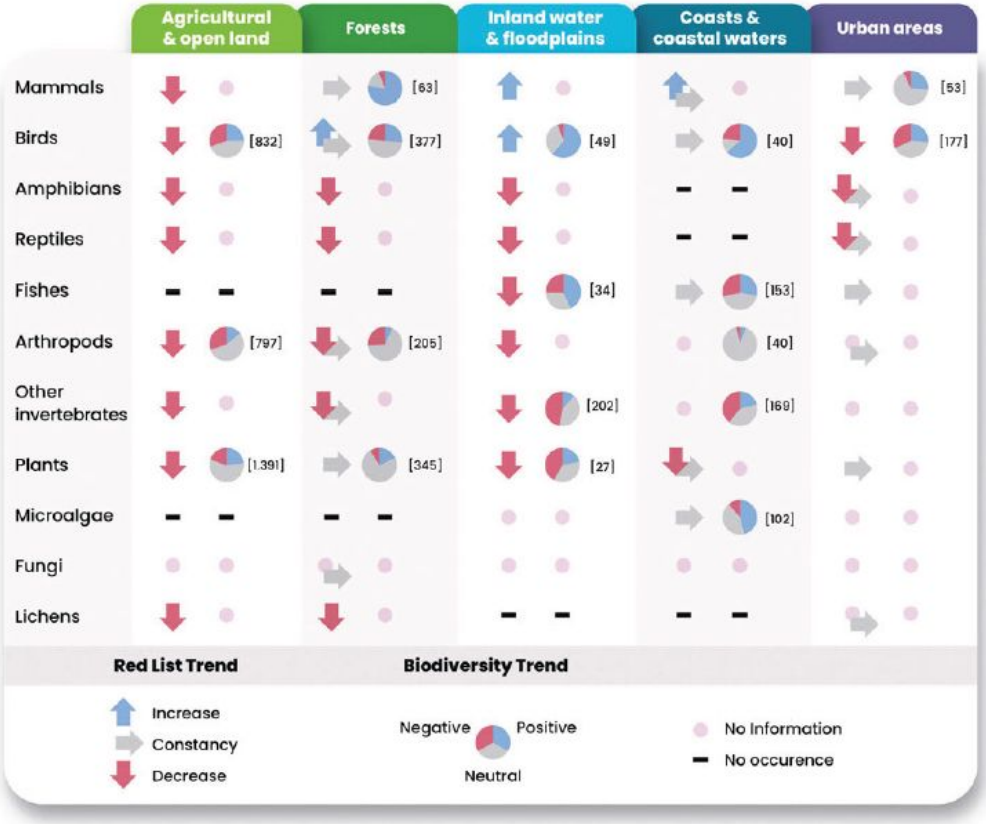
+

Lists of literature categorized as relevant / irrelevant

Literature database with annotation



List of biodiversity trends



Keeping the literature base up to date

Find new literature

Search for new biodiversity literature every two weeks

→

Screen for relevance

Build on the search queries used in Faktencheck

→

Support assessment

Provide full texts for extraction and expert review

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Overlap with the original Faktencheck literature

95% of the literature retrieved

97% of literature classified as relevant

Internal evaluation on original forest bibliography lists.

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WIP: semantic filtering learns from experts' relevance decisions.

Internal evaluation on original forest bibliography lists.

Buried information: What do we want to extract?

> The Alpine Ring Ouzel *Turdus torquatus alpestris* is a typical mountain bird that has recently declined in most of its breeding range. In this study we compared the historical (1986–1987) and recent (2017) distribution of breeding Ring Ouzels in 62 randomly selected 750 × 750 -m plots and analysed local colonization and extinction patterns in 558 subunits (= ‘grid cells’) in the southern Black Forest (south-western Germany). Our study revealed that habitat occupancy has decreased by about one-third, mainly at lower elevations and in depressed landforms, during the past three decades.

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Taxon · English	Taxon · scientific	Time	Location	Trend relation	…
Ring Ouzel	T. torquatus alpestris	1986–1987 2017	Black Forest Germany	(birds, forest, abundance, negative)	

Illustrative output from the broader extraction schema

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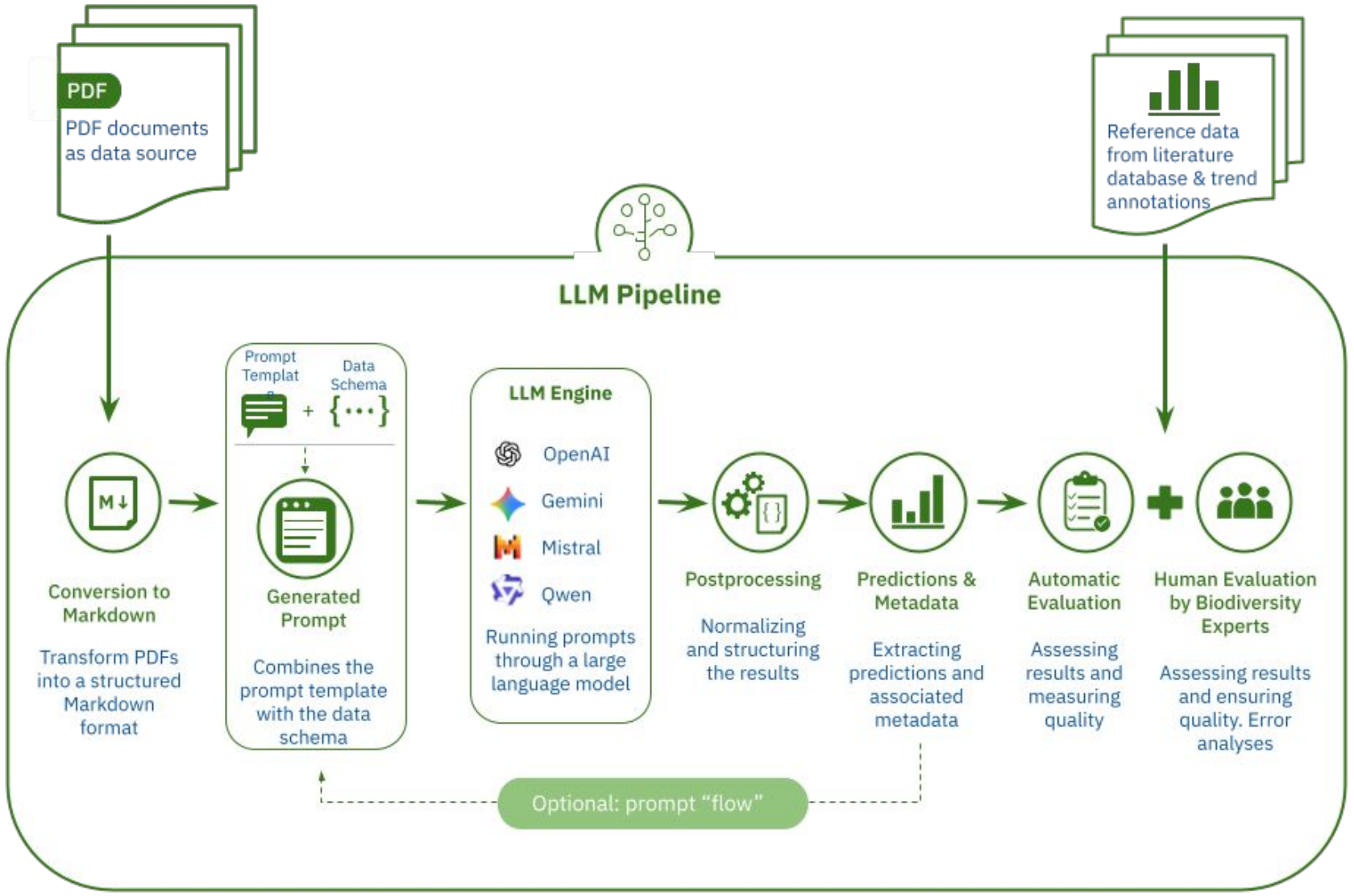
Taxon English · Taxon Scientific · Study Type · Time range · Time range · Biodiversity Variable · Spatial Measurements · Spatial Extent / Resolution · Spatial Extent / Resolution · Location · Location · Biodiversity Trend · Biodiversity Trend

Values come with traceable evidence.

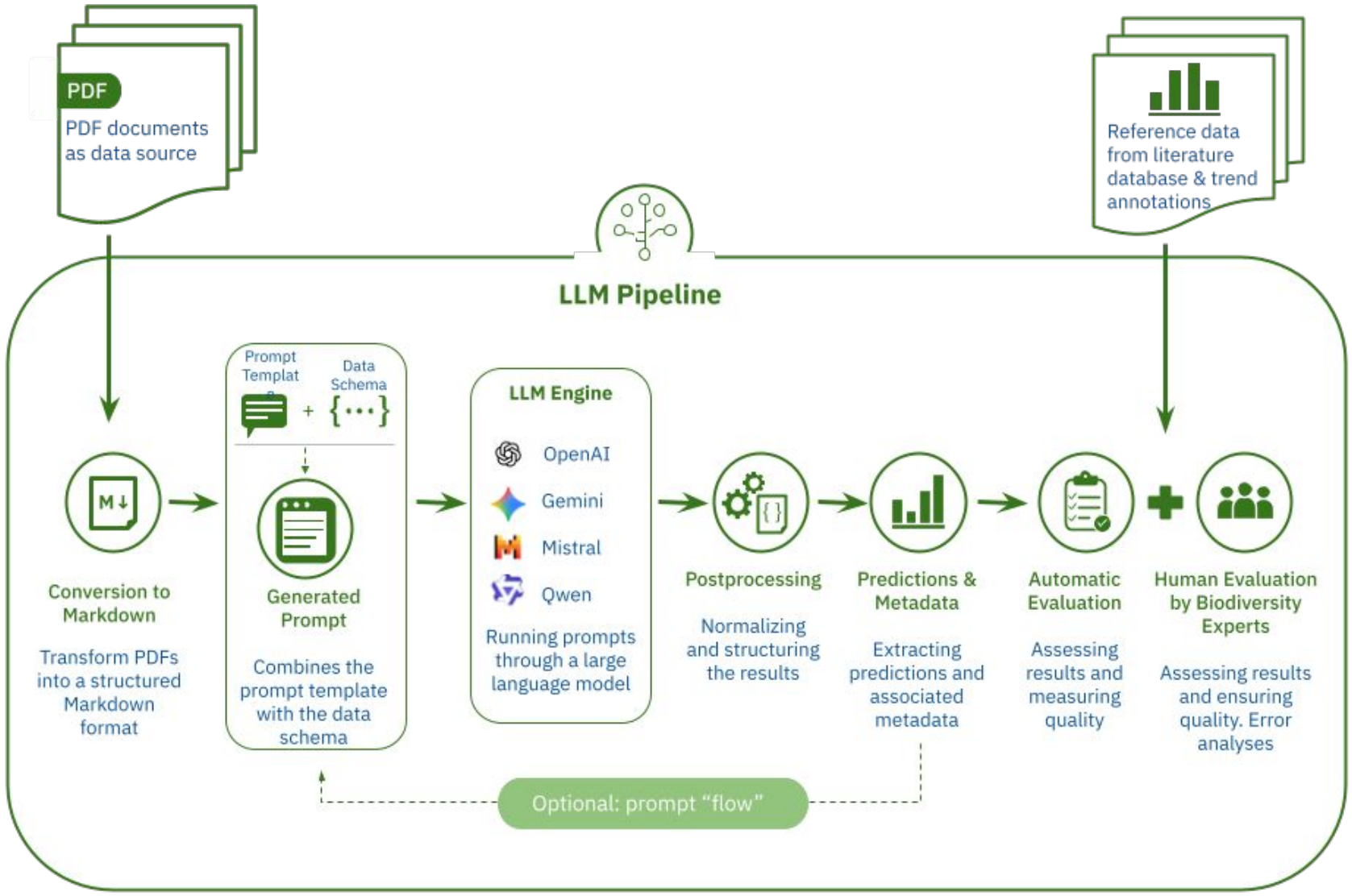
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Illustrative output from the broader extraction schema

Pipeline: document wise, schema guided extraction



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Details

- **Data schema**: model all data in one place
- **Guided decoding**: response is always valid JSON following the schema
- **Prompt** = template + generated schema description
- **Evidenz anchor**: provide extraction provenenance
- **Chunking** (20.000 chars): context limit and „lost in the middle“
- vLLM (lokal) or OpenAI-API; LlamaIndex, Hydra, Pydantic
- PDF → Markdown (PyMuPDF4LLM), full text incl. appendices

Two tasks, evaluated on real biodiversity literature

≈ 1,500 document inputs across development and testing

Task 1: What is the study about?

Classification as in the literature database, independent information

- Habitat
- Species group
- Biodiversity level
- Ecosystem type / category
- Ecosystem type / term
- ...

Test: 500 documents

Task 2: Which trend is reported?

All four parts must fit together

- Habitat
- Species group
- Variable
- Direction

Test: 430 documents
agricultural & open land, with and without trends

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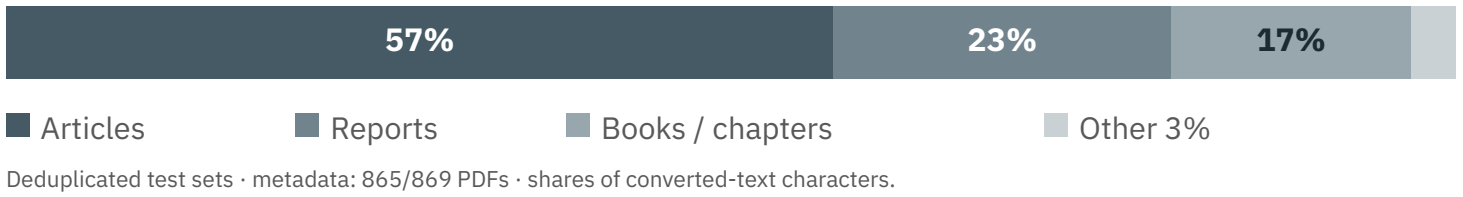
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Corpus characteristics

German & English; **21–37% in German**

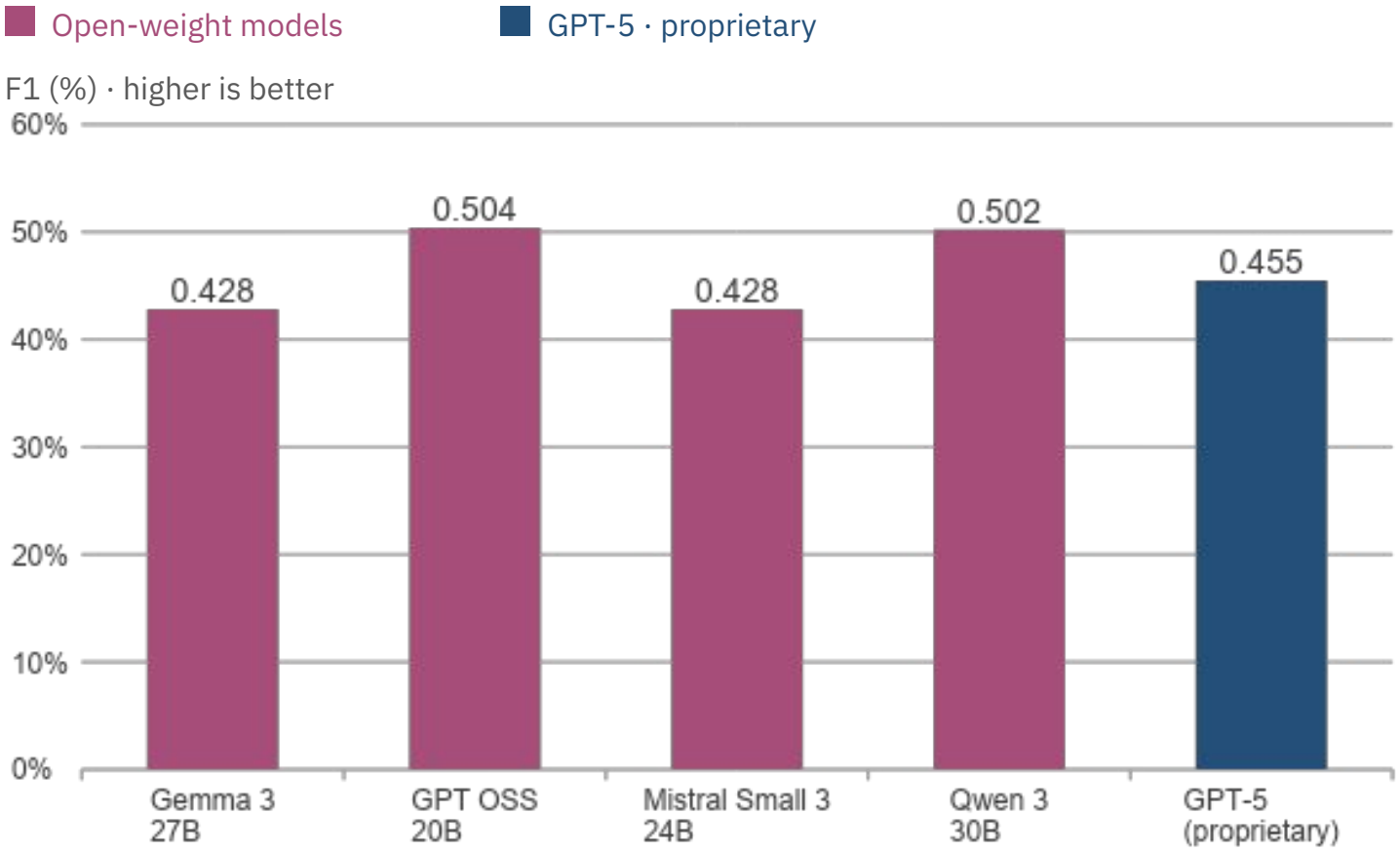
Up to 2,218 pages in a single book

Source types



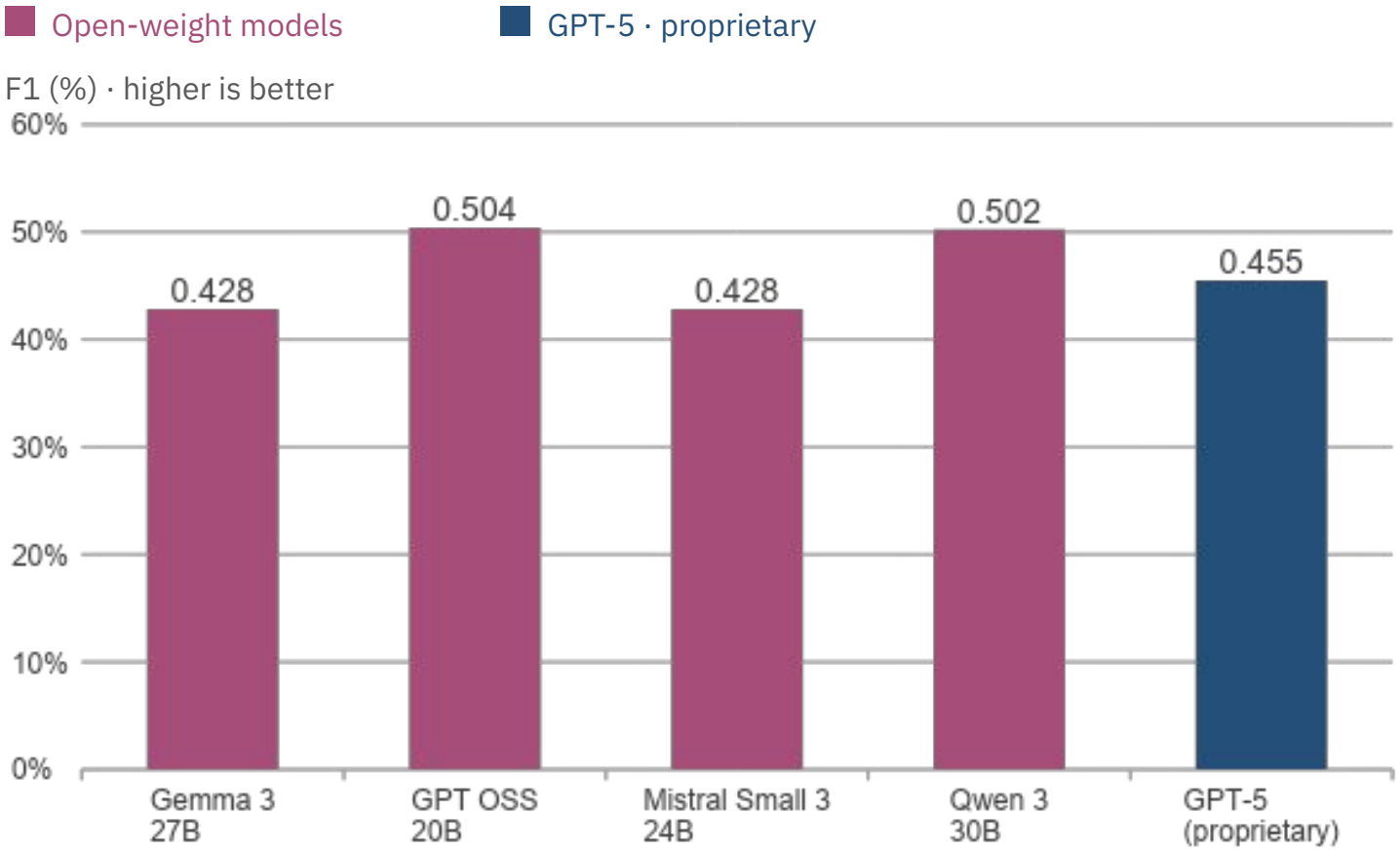
Deduplicated test sets · metadata: 865/869 PDFs · shares of converted-text characters.

Result 1: open-weight models reach about 50% F1



F1 balances finding reference information with avoiding unmatched answers.
100% means complete agreement with the reference.

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≈50%

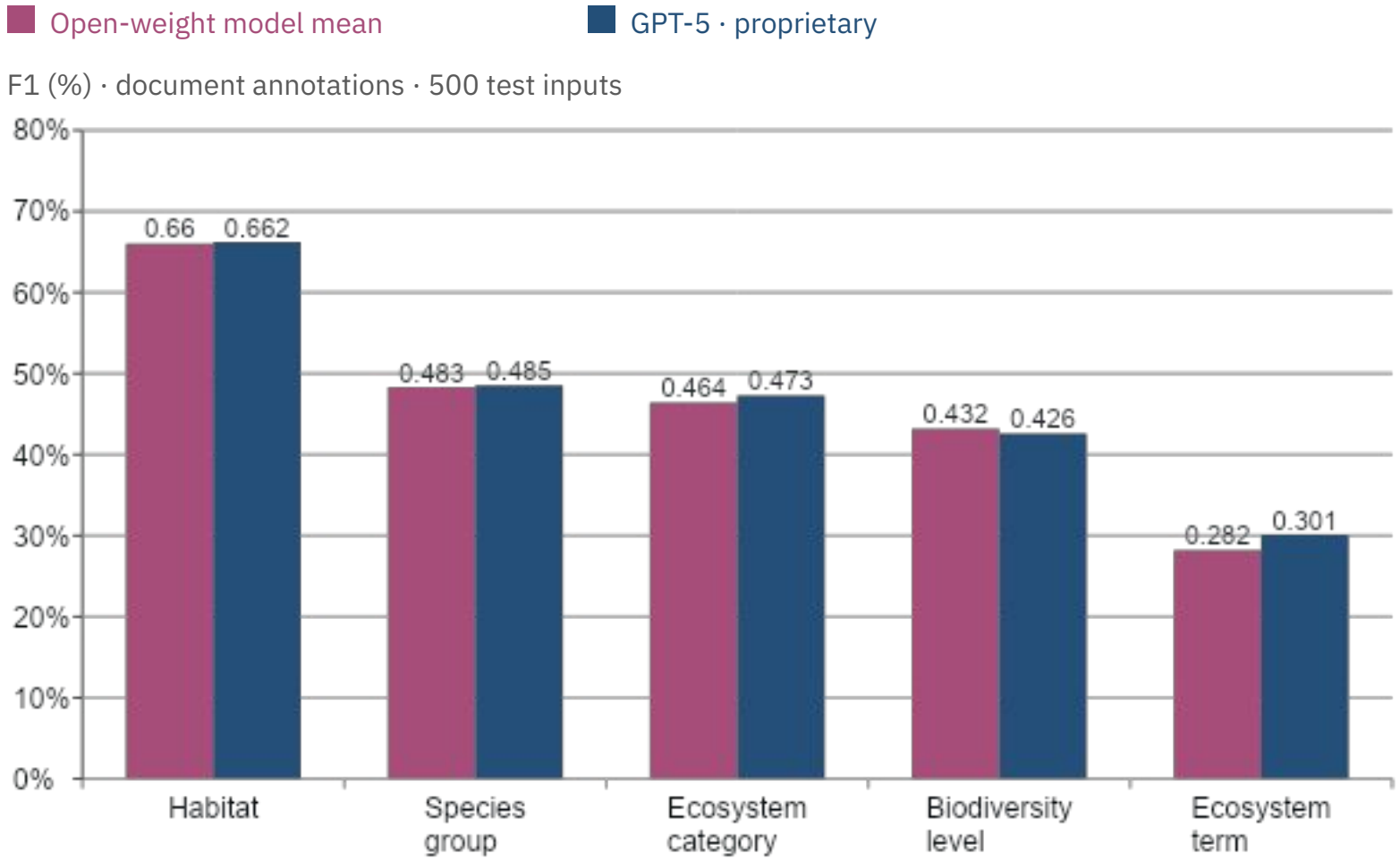
F1 for document annotations on 500 unseen documents

open-weight ≈ GPT-5

Best open model: 50.4% F1
GPT-5: 45.5% F1

F1 balances finding reference information with avoiding unmatched answers.
100% means complete agreement with the reference.

A closer look: AI is not equally good for all topics



Performance varies by annotation type.

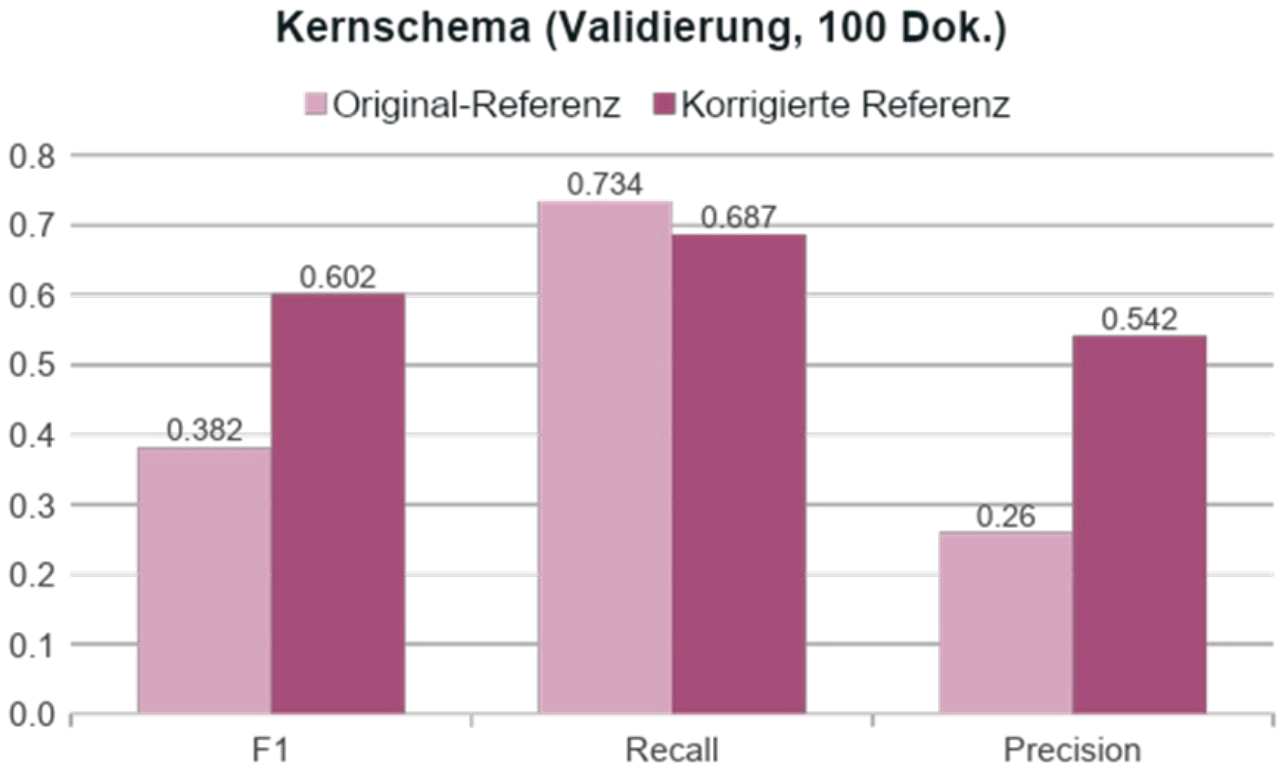
Habitat is easier to identify than detailed ecosystem terms.

Manuscript Fig. 6a · open-weight mean across four models, each averaged over three runs; GPT-5: one run.

Result 2: the AI finds most reference information – and more

74%

completeness (recall): about three quarters of the experts' annotations are found



Test set (500 docs), mean of open models; re-check: validation set (100 docs)

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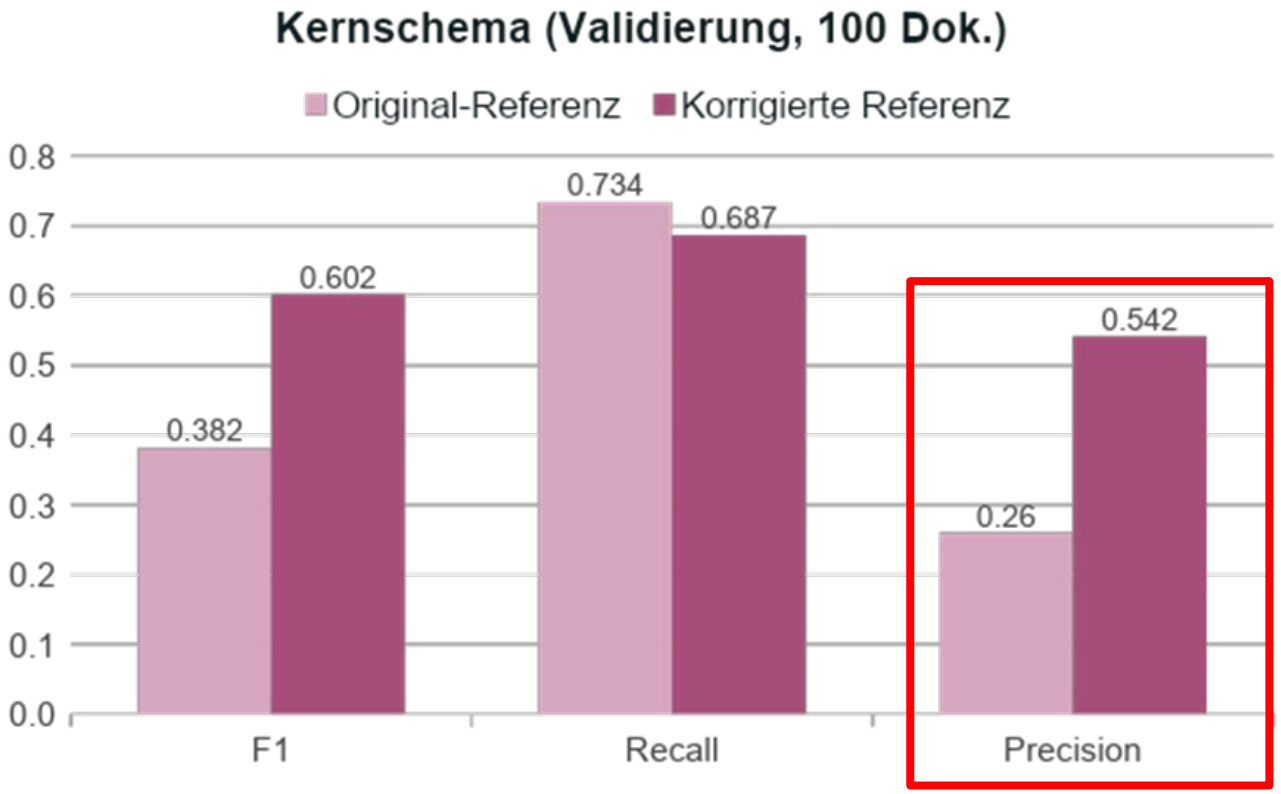
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34%

precision: many additional answers – real errors, or gaps in the reference?

410 → 914

annotations after correcting the reference



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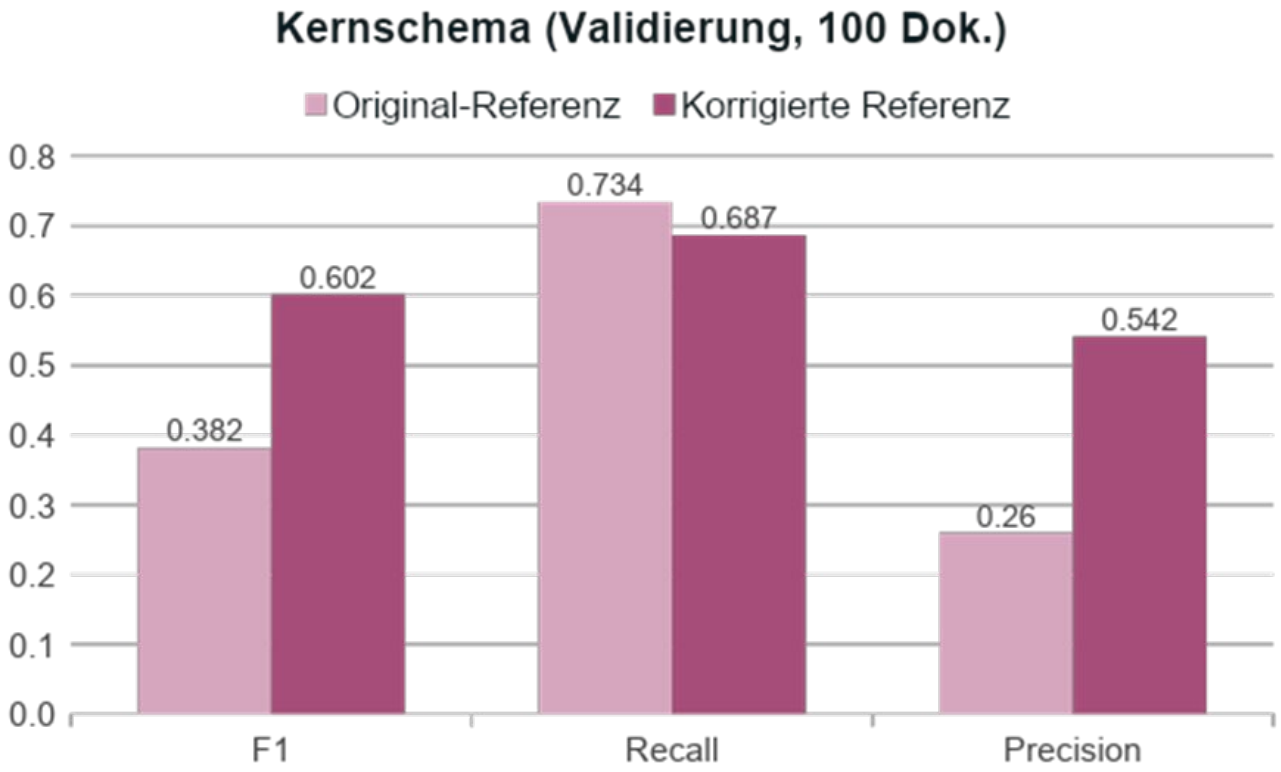
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Especially in long texts, AI can find information missed by human annotators.

Test set (500 docs), mean of open models; re-check: validation set (100 docs)

Where do LLMs go wrong? - Annotations (NER)



15% Missed information (False Negative I)



13% Technical failure (False Negative II)



5% Irrelevant information (False Positive I)



4% Misinterpretation (False Positive II)

Manual error analysis on validation data, share of all predictions; manuscript Tab. 3

Where do LLMs go wrong? - Annotations (NER)

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- different topics in one study → LLM focuses on one

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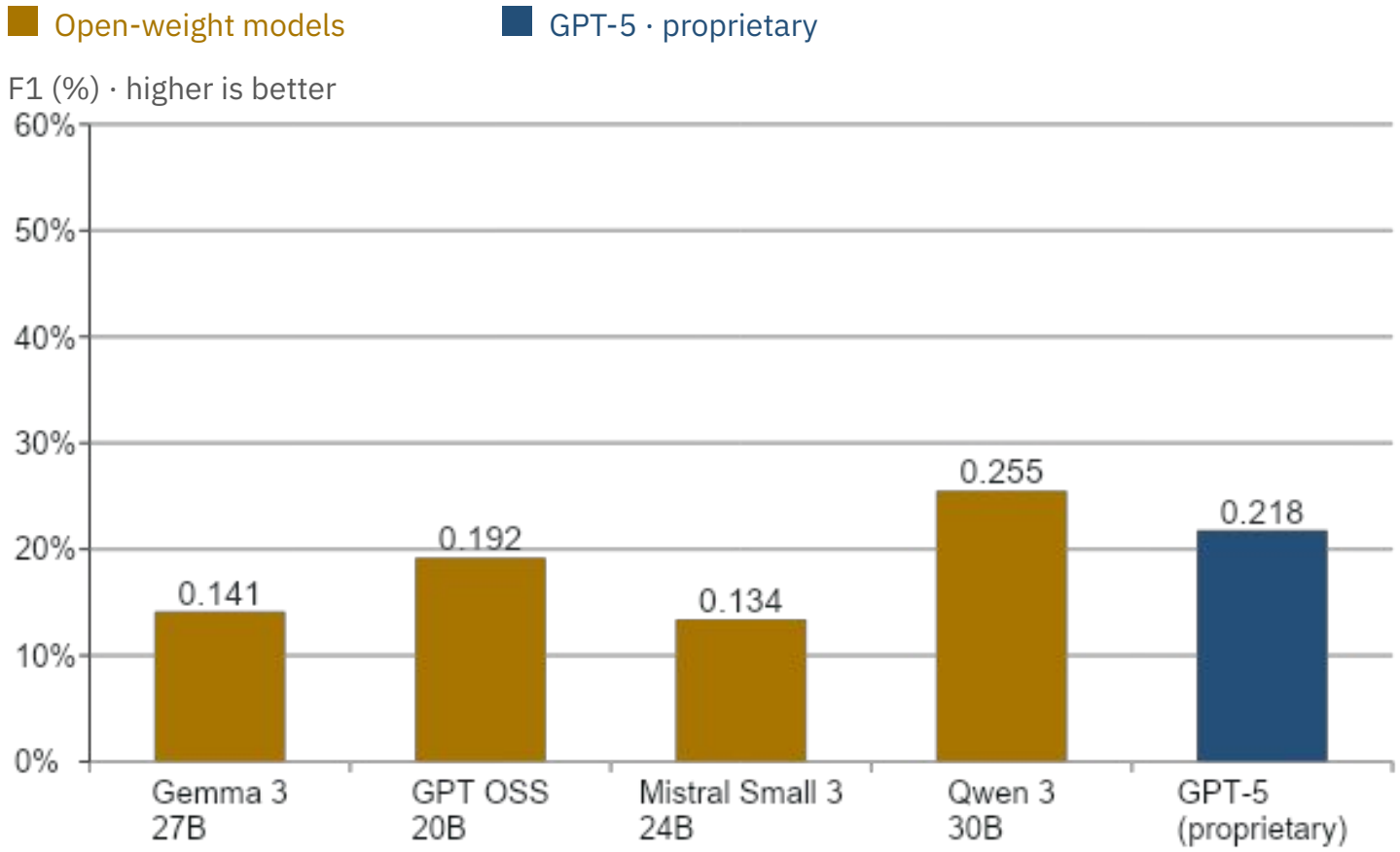
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- Assignments to wrong categories,
  e.g. “structure” too often assigned to “structural diversity”

Manual error analysis on validation data, share of all predictions; manuscript Tab. 3

Result 3: complete trend extraction remains difficult

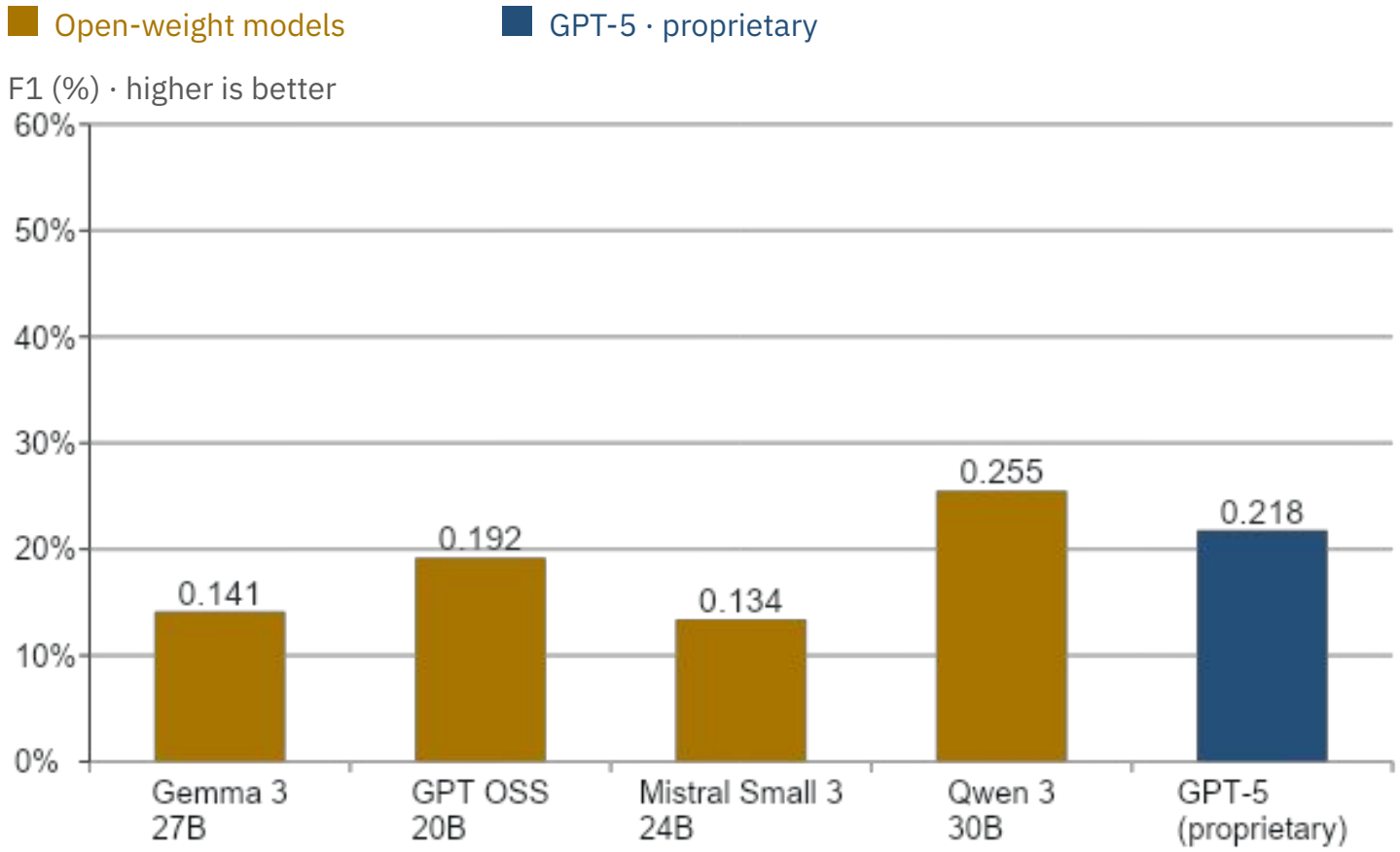


Habitat	Species group	Variable	Direction
forest	*birds*	*abundance*	*negative*

GPT-5 has the highest recall; Qwen has the highest precision and F1.

430 agricultural/open-land test inputs · exact match of habitat + species group + variable + direction · manuscript Fig. 4b.

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25.5%

Best F1 · Qwen 3 30B
Versus about 50% F1
for document annotations

Qwen achieved the highest observed F1 (GPT-5: 21.8%).

Where do LLMs go wrong? - Biodiversity trends (RE)



16%

Missed information
(False Negative I)



10%

Technical failure
(False Negative II)



21%

Irrelevant information
(False Positive I)



3%

Misinterpretation
(False Positive II)

Why is this so challenging?

- Four pieces of information, often scattered across the text (“long distance relations”)
- Reference relies on tables/ figures - LLM on text
- Language style, e.g. hedged wording, diversity of phrasings

Manual error analysis on validation data, share of all predictions; manuscript Tab. 3

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lower

decrease

declined

disappeared

Manual error analysis on validation data, share of all predictions; manuscript Tab. 3

What does large-scale extraction cost?

Resource	Per document	Full corpus · 3,917 PDFs
Open-weight models: GPU time	40–150 s	84 GPU hours
Open-weight models: hosting estimate	**≈€0.06**	**≈€235**

Approximate estimates for one run; representative configuration. Rental equivalent: €2.80/H100-hour excl. VAT (OVHcloud, 30 Oct 2026).
The runtime range spans models/tasks; 84 hours is one configuration. Costs vary by model, text length and provider.

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GPT-5: estimated API expenditure	**€0.08**	**€313**

Both incur costs: GPU hosting or usage-based API charges.

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Key findings

1. LLMs can analyse biodiversity literature at scale – zero-shot, without training, with open models.

2. LLMs reach performances of 50% for simple annotations (NER), 25% for more complex trend extraction (RE).

   With a similar reference-correction effect, F1 could reach ≈80% for NER.

3. LLMs make mistakes. So do human researchers.

(How) can LLMs support the literature work for assessments?

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In all options: experts stay in the loop – the AI provides evidence, humans judge and interpret.

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In parallel

AI and humans both annotate; differences are checked

+ maximum completeness

– no ressources saved

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AI takes on long texts such as books and reports; humans the rest

+ time saved where it costs most

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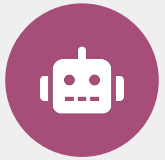
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AI takes on long texts such as books and reports; humans the rest

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– some risk of errors



AI first, human check

AI analyses everything; experts check random samples and estimate uncertainty

+ maximum time saved
– increased risk of errors

In all options: experts stay in the loop – the AI provides evidence, humans judge and interpret.

Outlook

Next steps

- Continuous error analyses & prompt improvements
- Semantic filtering for literature crawling
- Relation extraction for biodiversity and ecosystem services
- Digitize old reports

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Planned experiments

- Multi-query workflows & LLM-as-a-judge
- Domain-adapted models
- Harmonise species names (e.g. via GBIF)

Thank you!

KIBA-D Team

Leonhard Hennig, Diego Arenas, Arne Binder, David Harbecke •
Julia S. Ellerbrok, Annalena Lenk, Tobias Proß, Jori Maylin Marx, Maria Sporbert •
Luisa Schneider, Peter Gailhofer •
Helge Bruelheide, Nina Farwig, Josef Settele, Christian Wirth

Manuscript

„AI meets Biodiversity: How Large Language Models can Support Continuous Biodiversity Assessments“ (in preparation)

Code

github.com/DFKI-NLP/kibad-llm

With funding from the:

Federal Ministry
of Research, Technology
and Space



Backup

Where does complete trend extraction become difficult?

Qwen 3 30B · same extracted outputs, three evaluation views

Detecting trend statements

37.3 ± 1.1%

Habitat + species group + variable
Missed and extra combinations count as errors.

Classify the trend **direction**

66.9 ± 1.5%

Only correctly matched combinations
Missed and extra combinations are excluded.

Complete four-field extraction: 25.5 ± 1.2% F1

Trend detection remains difficult; stable and reversing trends are also less reliably recovered.

Matched-combination recall: declines 97%, increases 82%, neutral 52%, reversals 29–39% (few examples).

F1: mean ± sample SD over three seeds · diagnostic views; do not add or multiply the scores.

Which trend categories does Qwen recover?

Trend classification on correctly matched combinations · Qwen 3 · three seeds

Alignment-averaged, multi-label · matched targets only

Reference ↓ Predicted →	Negative	Neutral	Positive	Positive → negative	Negative → positive	Missing direction
Negative	**62.67**	0.00	0.44	0.00	0.00	1.22
Neutral	2.44	**15.33**	3.56	0.00	0.00	8.33
Positive	1.11	1.00	**30.00**	0.28	0.00	3.94
Positive → negative	0.67	0.17	0.00	**1.00**	0.00	0.83
Negative → positive	1.11	0.00	0.94	0.00	**2.00**	2.94
Unassigned prediction	25.67	13.17	28.06	1.06	1.00	0.00

Mean counts per seed; off-diagonals are expected error allocations.
Missing / unassigned labels occur within matched triples.

Exact per-direction diagnostics

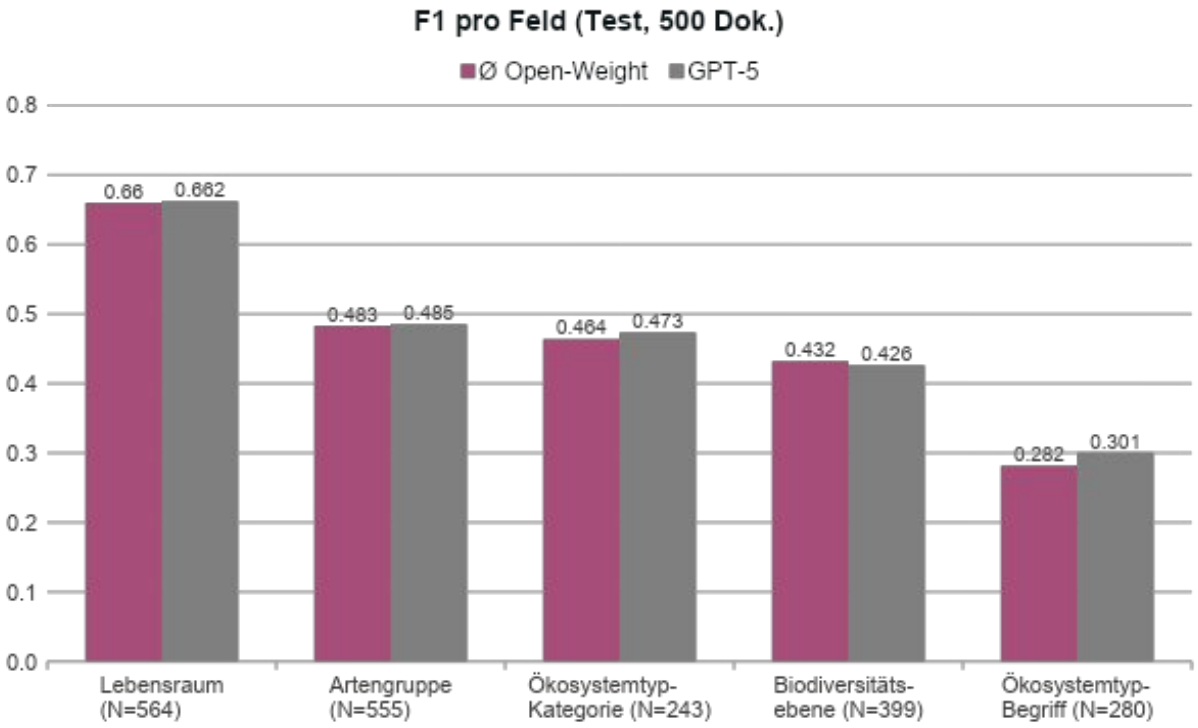
Direction	Precision	Recall	Ref. labels per seed
Negative	66.9%	97.4%	64–65
Neutral	52.5%	51.5%	29–31
Positive	47.6%	82.5%	35–39
Positive → negative	44.4%	38.9%	2–3
Negative → positive	66.7%	28.6%	7

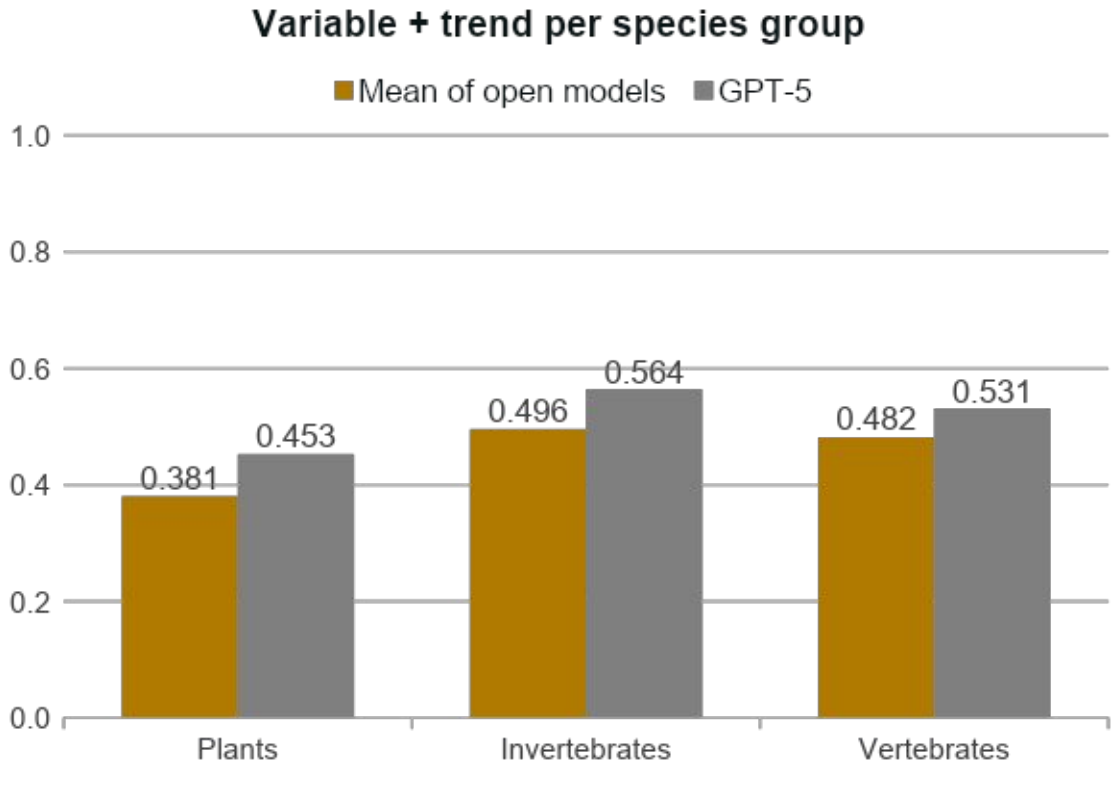
Few reference examples for reversals.
Precision / recall: means of per-seed ratios.

Qwen recovers increases and declines more reliably than stability or trend reversals.

Matched-combination recall: declines 97%, increases 82%, neutral 52%, reversals 29–39%. Conditional F1: 66.9 ± 1.5%.

A closer look: AI is not equally good for all topics





Beware of bias: information extraction does not work equally good for all topics.

Legal and governance questions



Copyright & text and data mining

Is mining allowed? Opt-out clauses (§ 44b German Copyright Act), licences, paywalls of sources



EU AI Act

Obligations for developing and using the models; documentation and transparency



Data governance

Roles and responsibilities for data quality and compliance; provenance for every extracted item



Re-use of outputs

Licensing of extracted data, correct attribution of sources, liability

Legal assessment and data governance concept by UfU and Öko-Institut (work in progress)